

Furthest Building You Can Reach

There are n buildings on a street, numbered from 1 to n , and the i -th building has height h_i . You start on the rooftop of building 1, and move from one rooftop to the next with the assistance of some ladders and bricks. You start out with l ladders and b bricks.

Suppose you are on the rooftop of the i -th building. If $h_i < h_{i+1}$, then you can move to the $(i+1)$ -th building by consuming either 1 ladder or $h_{i+1} - h_i$ bricks. If $h_i \geq h_{i+1}$, then you can move to the next building without consuming any ladders or bricks.

You are given n , the array $\mathbf{h}$, and the integers l and b . Compute the number of the furthest building you can reach if you use your ladders and bricks optimally.

Example Test Cases:

Test Case 1

Input: $n = 7$, $\mathbf{h} = [4, 2, 7, 6, 9, 14, 12]$, $l = 1$, $b = 5$

Output: 5

Explanation: You move from building 1 to 2 and 3 to 4 without any assistance. You consume $7 - 2 = 5$ bricks to go from building 2 to 3, and one ladder to go from building 4 to 5. You cannot move to building 6 because you have run out of ladders and bricks.

Test Case 2

Input: $n = 9$, $\mathbf{h} = [4, 12, 2, 7, 3, 18, 20, 3, 19]$, $l = 2$, $b = 10$

Output: 8

Explanation: You move from building 2 to 3, 4 to 5, and 7 to 8 without any assistance. You consume a ladder to go from building 1 to 2, $7 - 2 = 5$ bricks to go from building 3 to 4, a ladder to go from building 5 to 6, and $20 - 18 = 2$ bricks to go from building 6 to 7. You cannot move to building 9 because it requires 16 bricks, and you only have 3 left.

Expected Time Complexity: $O(n \log n)$

Input Format:

An integer n , an array $\mathbf{h}$ of length n , and two integers l and b .

Output Format:

Return a single integer, the number of the furthest building reached.

Constraints:

1. $1 \leq n \leq 10^5$

2. $1 \leq h_i \leq 10^6$

3. $0 \leq l \leq n$

4. $0 \leq b \leq 10^9$

Instructions:

You have been provided with two template files, containing the `main` and `solve` functions respectively. The `main` function, which has been completed for you, parses the input for each test case. For each test case, it calls the `solve` function, which must compute the number of the furthest reachable building.

The `solve` function must compute this in $O(n \log n)$ time.

(Refer to the `Instructions.pdf` file for detailed guidelines on the input-output format and additional test cases.)

Library Usage

For this lab, you are permitted to use any modules which are part of the standard library of either `C++` or `python`. For `C++`, do not add any extra `#include` directives; the provided `bits/stdc++.h` covers all allowed modules.